Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

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Abstract*—Multimodal Internet-of-Things deployments in clinical settings generate heterogeneous data streams that cannot be centralised, because the records are governed by consent and by data-protection regimes that prohibit pooling. Federated learning removes that obstacle by communicating model updates rather than records, and a hierarchical organisation relieves the wide-area link that otherwise constrains it; neither, however, supplies a privacy guarantee, and the guarantee such systems do report is conventionally stated over training records while the guarantee a clinical setting requires concerns individuals. Where a single episode of care yields many records, the two differ for every participant. This paper presents a three-tier hierarchical federated framework in which the privacy unit is the patient by construction: gradients are grouped by individual before they are bounded, the noise scale is calibrated against every client simultaneously so that the interposition of the edge tier is reflected in the reported budget, and the cost of releasing a conformal calibration quantile is composed into that budget rather than omitted from it. Representation learning is separated from private adaptation, with encoders pretrained on a population disjoint from every client and frozen, so that only a compact fusion head is privatised. The framework is evaluated on a cohort of surgical cases in which both modalities originate from the same patient during the same operation, and the resulting guarantee is audited by mounting membership inference at the level of the individual, with every attack reported alongside a positive control on a configuration known to memorise. A further experiment shows that the price of the individual-level guarantee is not a fixed quantity but a function of how many records each individual contributes, vanishing when that number is one — where the two units coincide by construction, so that the experiment controls itself — and growing monotonically thereafter. [Headline quantitative findings — uplink reduction, privacy budget, inference latency, energy, and the membership-inference audit — to be inserted once Results is complete.]

***Index Terms*—Hierarchical federated learning, Internet of Things, differential privacy, user-level privacy, membership inference, multimodal learning, edge computing, quality of service, conformal prediction.**

I. Introduction

Internet-of-Things deployments increasingly derive their value from combining sensing streams of different kinds. A hospital theatre records continuous physiological waveforms alongside the structured clinical history of the patient on the table; a wearable installation records inertial motion alongside egocentric video. In each case the streams are complementary rather than redundant, and a model with access to both can resolve states that neither resolves alone. The streams are also produced where the sensors are, which is seldom where computation is convenient, and this tension between where data arises and where models are trained is the setting the present work addresses.

Transmitting raw multimodal streams to a central service is the obvious resolution and is frequently unavailable. The volume is substantial and the uplink is the scarce resource, so centralisation inflates both latency and energy in precisely the deployments that can least afford either. More decisively, in the settings where multimodal sensing is most valuable the data cannot lawfully move at all: clinical records are governed by consent and by data-protection regimes that prohibit pooling across institutions, so a model requiring centralisation cannot be deployed rather than merely being expensive. Federated learning [1] answers this by communicating model updates instead of records, and hierarchical variants [2], [3] interpose an edge tier so that the wide-area link is exercised once per several local rounds.

Moving the model rather than the data does not by itself protect the data. The updates a client transmits are functions of its records, and membership inference against released models is well established [4]. A formal guarantee is therefore required, and differentially private stochastic gradient descent [5] supplies one by bounding the influence of a training record and perturbing the aggregate. But the guarantee such a system reports is stated over the unit it clips, and in a sensing deployment that unit is a fixed-length window rather than a person. Where an individual contributes many windows — as every patient does, and as every participant in a continuous recording does — a record-level budget answers whether a particular window was used in training, while the question a clinical deployment must answer is whether a particular *person* was. The two can be related by group privacy [6], but at the group sizes these datasets exhibit the conversion is prohibitive, turning a nominal budget of unity into one that supports no meaningful claim.

Existing systems have developed along two lines that do not meet. Hierarchical designs establish the communication and energy case convincingly: HED-FL [3] extends deployment lifetime by avoiding long-range transmission, HCEF [7] adapts computation and compression to device heterogeneity, and assignment-oriented work [8] reduces divergence under non-

identical distributions. None carries a formal privacy guarantee. Private designs supply the guarantee: FL2DP [9] couples noise with shuffling, and RoPPFL [10] pairs local privacy with robust aggregation at the edge. Neither is hierarchical in the sense that matters here, and neither examines how an intermediate tier alters the frequency with which each client's budget is spent. Across both lines the local objective is treated as the aggregation of generic parameter vectors, so neither multiple modalities per record nor the grouping of records under an individual plays any part.

Three difficulties follow, and they bear on deployability as much as on privacy. The unit at which a guarantee is stated is rarely examined, so its cost on genuinely paired multimodal data is unquantified. The validity condition of parameter-efficient private training — that the corpus used to pretrain the frozen representation be disjoint from the protected population [11], [12] — is frequently left implicit and, where honoured, unpriced. And guarantees are typically reported from the parameters of the mechanism rather than audited by attack; where an audit is performed it is conducted at the granularity of the record, which is precisely the quantity a record-level guarantee already protects.

Two remedies follow directly. The unit can be fixed at the point of clipping rather than recovered afterwards by a group-privacy conversion: if the gradients of an individual's records are accumulated before the bound is applied, that individual's influence on the released model is bounded irrespective of how many records they contributed, and accounting performed at the same granularity yields a budget that is a statement about people by construction. And the resulting guarantee can be audited by mounting membership inference at the level of the individual, provided the attack used is strong enough that a negative finding is informative — which requires demonstrating on a configuration known to memorise that the attack can find leakage where leakage exists.

Accordingly, this work presents PP-HFM, a three-tier federated framework for multimodal sensing in which the privacy unit is the individual by construction. Records are grouped by individual before they are bounded; the noise scale is calibrated to satisfy the budget for every client simultaneously rather than for a representative one, so that the interposition of the edge tier is reflected in the reported figure; and the cost of releasing a conformal calibration quantile is composed into that figure rather than omitted from it. Representation learning is separated from private adaptation, with encoders pretrained on a population disjoint from every client and frozen, so that only a compact fusion head is privatised. The framework is evaluated on two corpora whose modalities are associated by the recording apparatus rather than by a construction rule, under a protocol that measures communication, latency, reliability and energy on a simulated topology and audits the privacy guarantee by attack.

The contributions are fivefold. First, the framework enforces an individual-level guarantee within a hierarchical multimodal system and reports a composed budget covering both training and calibration; the accounting practices this requires — calibrating the noise against every client rather than a representative one, and charging the calibration release against the same budget — are specified in enough detail to be checked from outside, because each corresponds to a failure that is invisible in a training log. Second, the cost of the choice of unit is measured on genuinely paired clinical multimodal data, together with the cost of the disjointness condition that makes the guarantee valid end-to-end, so that both are priced rather than assumed negligible. Third, the guarantee is audited by patient-level membership inference under both a threshold attack and a likelihood-ratio attack [13], each reported alongside a positive control on a configuration known to memorise; the audit returns a null on the private arms, and the control is what makes that null attributable to the defence rather than to an inadequate adversary. Fourth, the system-level behaviour is validated on two deployments differing in topology and in payload, with every network quantity measured on rounds in which the payload traversed the simulated topology rather than composed analytically. The same protocol establishes that the guarantees are affordable in real time: the conformal set construction that carries the coverage guarantee occupies $0.68\,\%$ of the inference path, and the private mechanism occupies none of it, since its noise is applied to gradients during training and adds no operation to serving. The objection that a privacy apparatus is too heavy for a latency-bound deployment is therefore answered by measurement rather than by argument. Fifth, the price of the individual-level guarantee is shown not to be a fixed quantity but a function of how much each individual contributes: with the per-step budget and the number of units per step held constant, the penalty is indistinguishable from zero when an individual contributes a single record and grows monotonically as that number rises, saturating by eight records. This design carries its own control, because at one record per individual the two units coincide by construction and the two arms must agree; that they do is what licenses reading the divergence at larger contributions as a property of the unit rather than of the apparatus.

What is not claimed should be stated as plainly. User-level differential privacy [14], parameter-efficient differentially private fine-tuning [11], [12] and differentially private quantile release [15] are established prior art, and are used here rather than proposed. The contributions above are measurements of these mechanisms in a setting where they have not been measured, and the auditing practice under which those measurements are reported.

The remainder of the paper is organised as follows. Section II reviews related work on hierarchical and private federated learning, on the unit at which a guarantee is stated, and on membership inference as an auditing instrument; Section III presents the proposed framework; Section IV specifies the corpora, the implementation, the evaluation measures and the experimental protocol; Section V reports the results; Section **??** discusses their interpretation and the limitations of the study; and Section **??** concludes.

II. Related Work

A. Federated Learning and Hierarchical Aggregation

The formulation of distributed training as the communication of model updates rather than of data [1] established

the basis upon which subsequent federated systems were constructed. In its flat form, however, every participating client transmits to a central server in every round, so that the wide-area link rather than local computation determines how frequently the global model may be revised. Client–edge–cloud designs [2] addressed this by interposing an intermediate tier at which several rounds of local optimisation are absorbed before one round of central aggregation is performed. HED-FL [3] developed the principle for constrained deployments, clustering devices dynamically at the edge and demonstrating that the avoidance of long-range transmission substantially extends the operational lifetime of an installation, while HCEF [7] introduced heterogeneity awareness, adjusting per-device computation frequency and compression ratio online so that the slowest participant does not dictate the duration of a round. Mhaisen *et al.* [8] treated the assignment of clients to aggregators as the object of optimisation, showing that an assignment informed by the statistical properties of local data reduces model divergence under non-identically distributed conditions. These works establish the communication and energy case for hierarchy convincingly, and the present study adopts rather than re-derives it. None of them, however, carries a formal privacy guarantee: HED-FL provides none, and the assignment strategy of [8] requires clients to disclose local label statistics, which is itself a release of information about the protected population and is difficult to reconcile with a strict budget.

B. Differential Privacy in Federated Learning

Differentially private stochastic gradient descent [5] supplies the guarantee that hierarchical designs omit, bounding the influence of a training record by clipping its gradient to a fixed norm and perturbing the aggregate with calibrated noise, with the accumulated cost tracked by a moments or Rényi accountant [16] within the framework of [6]. Within federated learning the mechanism has been developed in two principal directions. FL2DP [9] combines an exponential noise mechanism with a shuffling entity that severs the association between a client and the update it contributes, reporting improved accuracy at an equivalent formal guarantee, while RoPPFL [10] pairs local differential privacy with similarity-based robust aggregation performed at the edge, so that the intermediate tier serves as a verifier against corrupted updates as well as a relay. Both establish that a formal guarantee is attainable in the federated setting at acceptable cost. Both, however, are organised flatly with respect to the property considered here: FL2DP provides differential privacy without a hierarchy in the sense that the edge tier plays no part in the accounting, and neither examines how the interposition of such a tier alters the frequency with which each client's budget is expended, which is the quantity upon which the calibration of the noise ultimately depends.

C. The Unit to Which a Guarantee Attaches

The distinction between a guarantee stated over records and one stated over individuals is long established and is not a contribution of the present work. McMahan *et al.* [14] introduced user-level differential privacy for federated language models, bounding the contribution of each user rather than of each example, and the group-privacy relation by which one may be converted into the other is standard [6]. The mechanism is therefore available; what is not available is a measurement of what it costs in the setting that most requires it. In clinical monitoring the grouping of records under an individual is not a property of a few atypical participants but of every participant, since a single episode of care yields many time-aligned records, and the conversion factor is consequently applied to the whole population rather than to its tail. The literature supplies neither the utility forgone when the unit is moved from the record to the individual on genuinely paired multimodal clinical data, nor the manner in which that cost varies with the number of records an individual contributes, and in the absence of the latter a single measurement cannot be distinguished from a property of the particular cohort on which it was obtained.

D. Parameter-Efficient Private Adaptation

The perturbation that a parameter vector must absorb under a fixed budget grows with its dimensionality, so that private training of an entire network degrades severely at the budgets a clinical claim requires. Restricting the private optimisation to a compact trainable subset situated above frozen pretrained representations is an established remedy, and Li *et al.* [11] and Yu *et al.* [12] demonstrate that parameter-efficient differentially private fine-tuning is markedly more robust at small ε than end-to-end private training. The present work adopts this recipe and claims no novelty for it. The recipe nevertheless carries a precondition that is frequently left implicit: the corpus upon which the frozen representations are obtained must be disjoint from the protected population, since otherwise the representations have already been derived from that population without a guarantee and the budget subsequently reported for the trainable component does not describe the system as a whole. Where the precondition is honoured it is seldom priced, so the utility conceded in order to obtain a guarantee that is valid end-to-end remains unquantified; and the consequences of the recipe for what the resulting model retains of its training population have received little attention.

E. Membership Inference as Empirical Auditing

A differential privacy guarantee is an upper bound upon the influence of a participant and does not by itself establish what an adversary attains in practice, for which membership inference supplies the complementary measurement. Shokri *et al.* [4] introduced the shadow-model formulation, and thresholding of the loss or confidence [17] remains the most commonly reported attack. Such attacks are weak by construction: a single global threshold is confounded by the varying difficulty of individual examples, so that an easily fitted non-member may be scored above a poorly fitted member and the reported strength constitutes a lower bound upon leakage rather than an estimate of it. Watson *et al.* [18] demonstrated that calibrating against per-example difficulty alters conclusions materially, and Carlini *et al.* [13] formalised the calibration

as a likelihood-ratio test between per-example in- and out-distributions estimated from shadow models, arguing further that attacks should be assessed by their true-positive rate at low false-positive rate rather than by an average-case summary that conceals confident identification of a minority. Two limitations persist. Membership is almost invariably evaluated at the granularity of the training record, which is precisely the quantity that a record-level guarantee already protects and not the quantity a clinical guarantee concerns; and negative results are rarely accompanied by evidence that the attack employed was capable of a positive one, so that an absence of detected leakage cannot be distinguished from an inadequate adversary.

F. Multimodal and Clinical Federated Learning

The motivation for federated learning is particularly clear in medicine, where governance rather than volume prevents the assembly of data at a single site [19], [20]. The majority of federated work, including the systems discussed above [3], [7], [9], [10], nevertheless treats the local objective as the aggregation of generic parameter vectors and does not represent multiple modalities per record, notwithstanding that clinical evidence is characteristically multimodal and that the modalities available at one institution may differ from those available at another. Where multimodal federated benchmarks have been assembled, the modalities are frequently associated by a construction rule rather than by common provenance, so that an apparent benefit of fusion may reflect the regularity of the rule rather than genuinely complementary information; a benchmark of this kind cannot support a claim about fusion at all. Corpora in which both modalities originate from the same individual during the same episode and are aligned at the point of capture [21], or are synchronised in hardware [22], avoid the difficulty entirely, since the association is a property of the source and requires no defence.

G. Uncertainty Quantification under a Privacy Budget

Conformal prediction provides prediction sets carrying a distribution-free coverage guarantee and requires neither retraining nor distributional assumptions [23], [24], which recommends it for clinical deployment, where a point prediction unaccompanied by a statement of confidence is of limited use. Its application within a private system introduces a complication that is easily overlooked: the procedure requires the release of a quantile of nonconformity scores computed upon held-out data belonging to the protected population, and such a release is a disclosure in exactly the sense the surrounding framework exists to bound. Differentially private release of a quantile by the exponential mechanism is itself standard [15], so the mechanism is not at issue. What is at issue is the accounting: the cost of calibration composes with the cost of training, and a system that reports only the latter understates the budget it has expended, by a factor that grows with the number of occasions on which calibration is performed.

H. Real-Time Inference and Quality of Service at the Edge

A federated system is trained once and queried continuously, and the constraint that governs its clinical usefulness after deployment is the delay between a measurement arriving and a decision being available, rather than the time taken to reach convergence. The federated literature has addressed the latter extensively and the former hardly at all: HCEF [7] adapts computation frequency and compression ratio online, but the quantity it optimises is the duration of a training round; RoPPFL [10] places verification at the edge in order to reject corrupted updates, prioritising robustness over response time; and HED-FL [3] is concerned with the energy consumed in reaching a model rather than the latency of using one. Where inference cost has been considered it has generally been within the edge-computing literature rather than the federated one [25], and the principal instrument is reduced-precision arithmetic, integer quantisation in particular being well established as a means of lowering both memory footprint and inference time on commodity hardware [26]. The consequence is a gap that matters precisely for the systems this work targets: a federated design may satisfy a strict privacy budget and still be unusable at the bedside if the delay between observation and prediction exceeds the window in which a clinician can act, and no existing federated system reports an end-to-end inference latency measured under the privacy mechanism it deploys, alongside the communication and energy costs incurred in producing the model that serves it.

I. Summary and Research Gap

These lines of work converge upon the setting from which the present study proceeds, and five observations remain. First, hierarchy and formal privacy have been developed largely in separation — the hierarchical systems provide no guarantee, and the private systems do not examine how an intermediate tier alters the expenditure of the budget — so that the interaction between the two is unmeasured although each is individually well understood. Second, the mechanism for stating a guarantee over individuals rather than records is established, but its cost on genuinely paired clinical multimodal data is not, and in the absence of any characterisation of how that cost varies with the number of records an individual contributes, a single measurement cannot be distinguished from an artefact of the cohort upon which it was obtained. Third, parameter-efficient private adaptation is widely adopted, yet the disjointness of the pretraining corpus upon which its validity depends is frequently left implicit and, where honoured, is not priced. Fourth, guarantees are typically reported from the parameters of the mechanism rather than audited by attack; where an audit is conducted it is performed at the granularity of the record, and a null result is seldom accompanied by the evidence that would distinguish an effective defence from an inadequate adversary. Fifth, and cutting across the foregoing, the cost of a federated design is reported in the currency of training — rounds to convergence, bytes exchanged, energy expended — while the delay experienced by the clinician who queries the deployed model is left unmeasured, notwithstanding that it is this delay which determines whether the system can be used at the point of care at all.

The present work addresses these five points together. It combines a two-tier hierarchy with a guarantee whose sensi-

tivity is enforced at the level of the individual at the point of clipping rather than recovered by conversion afterwards, and calibrates the noise against every client simultaneously so that the interposition of the edge tier is reflected in the reported budget. It measures the utility conceded by moving the unit from the record to the individual, and characterises that cost as a function of the number of records each individual contributes rather than at a single operating point. It enforces and verifies the disjointness of the pretraining population, and reports the utility forgone in order that the resulting guarantee be valid end-to-end. Finally, it audits the guarantee by mounting membership inference at the level of the individual, using both a threshold attack and a likelihood-ratio attack calibrated per individual, and reports every result alongside a positive control on a configuration known to memorise, so that a negative finding may be attributed to the defence rather than to the adversary. Finally, it specifies and instruments the deployment path, so that the end-to-end latency of a prediction, the volume of traffic the hierarchy carries, and the energy each tier expends are reported for the same configuration that carries the privacy guarantee, rather than for a differently-configured system.

III. Methodology

A. Overall Framework of the Proposed PP-HFM

Multimodal Internet-of-Things deployments derive their value from combining sensing streams of different kinds — physiological monitors alongside clinical records, inertial sensors alongside video — and from doing so where the streams are produced rather than where a server happens to be. The evidence that would train such a model is therefore distributed by construction, and in the settings that matter most it cannot be centralised: records are governed by consent and by data-protection regimes that prohibit pooling, so a model requiring the data to move cannot be deployed at all. Federated learning removes that obstacle by moving the model instead of the data, but it does not by itself supply a privacy guarantee, because the updates a client transmits are themselves functions of its records. A framework intended for these settings must therefore combine distributed multimodal training with a formal guarantee, and must do so within the network, latency and energy budgets of an edge installation rather than those of a datacentre.

Differentially private federated learning supplies the formal guarantee, and the mechanisms for it are well established. What is not established is the *unit* to which that guarantee attaches in a clinical setting. Standard differentially private stochastic gradient descent [5] clips and accounts per training record, and in a monitoring dataset a record is a fixed-length window rather than a person: in the cohort used here every patient contributes several windows, so a record-level budget answers whether a particular window was used in training while the clinical guarantee must answer whether a particular *patient* was. Prior federated systems illustrate the same division from the opposite direction. HED-FL [3] establishes the communication and energy case for a hierarchical tier but carries no differential privacy; FL2DP [9] provides differential privacy without a hierarchy; and neither, nor the heterogeneity-aware and assignment-oriented designs that follow them [7], [8], models multiple modalities per record or the grouping of records under a single individual.

Accordingly, this work pursues a framework in which the privacy unit is the patient by construction rather than by conversion after the fact, and in which the hierarchy that makes the system deployable and the guarantee that makes it publishable are designed together rather than adopted separately. The system, termed PP-HFM (Patient-Private Hierarchical Federated Multimodal learning), is organised in three tiers: institutional clients that hold patient-grouped multimodal records and never release them, edge aggregators that absorb several local rounds before communicating upward, and a cloud aggregator that maintains the global model. Within each client, representation learning and private adaptation are separated: encoders pretrained on a disjoint public population are frozen, and only a compact fusion head is trained under a differentially private objective whose sensitivity is enforced at the level of the patient. Fig. 1 presents the overall organisation.

The processing pipeline proceeds as follows. Each client draws a batch of patients rather than a batch of windows, accumulates the gradients of all windows belonging to a sampled patient into a single per-patient gradient, bounds that gradient, and perturbs the aggregate once per optimisation step, so that the influence of any one individual on the released model is bounded irrespective of how many records that individual contributed. The resulting update is compressed and transmitted to the client's edge, which averages over its clients; after several such edge rounds the edges report to the cloud, which averages over edge populations and redistributes the global model. At the end of training the clients release differentially private calibration quantiles, from which a distribution-free prediction set is formed, and the privacy cost of that release is charged against the same budget as training.

The components are stated here without implementation detail. The patient-level privacy mechanism enforces sensitivity with respect to the individual rather than the record; the multi-client noise calibration ensures that a single noise scale satisfies the budget for every client simultaneously despite pronounced differences in client size; the two-tier aggregation converts local optimisation into a global model while exercising the wide-area link once per several local rounds; the update encoding reduces the transmitted payload; and the federated conformal calibration supplies coverage guarantees under the same budget. The deployment path specifies how a released model is executed within a latency budget, and the network and energy instrumentation measures the communication and energy costs the hierarchy incurs. The empirical audit that tests the resulting guarantee is described alongside them. The encoder architecture, the aggregation rule, and the conformal machinery are adopted from established work and are not claimed as contributions.

Taken together, these components constitute a framework in which a clinical privacy guarantee is stated at the unit clinicians care about, the cost of stating it that way is measured rather than assumed, and the guarantee is audited by attack rather than asserted from its parameters. The remaining subsections describe each element in detail, beginning with the architecture that organises them.

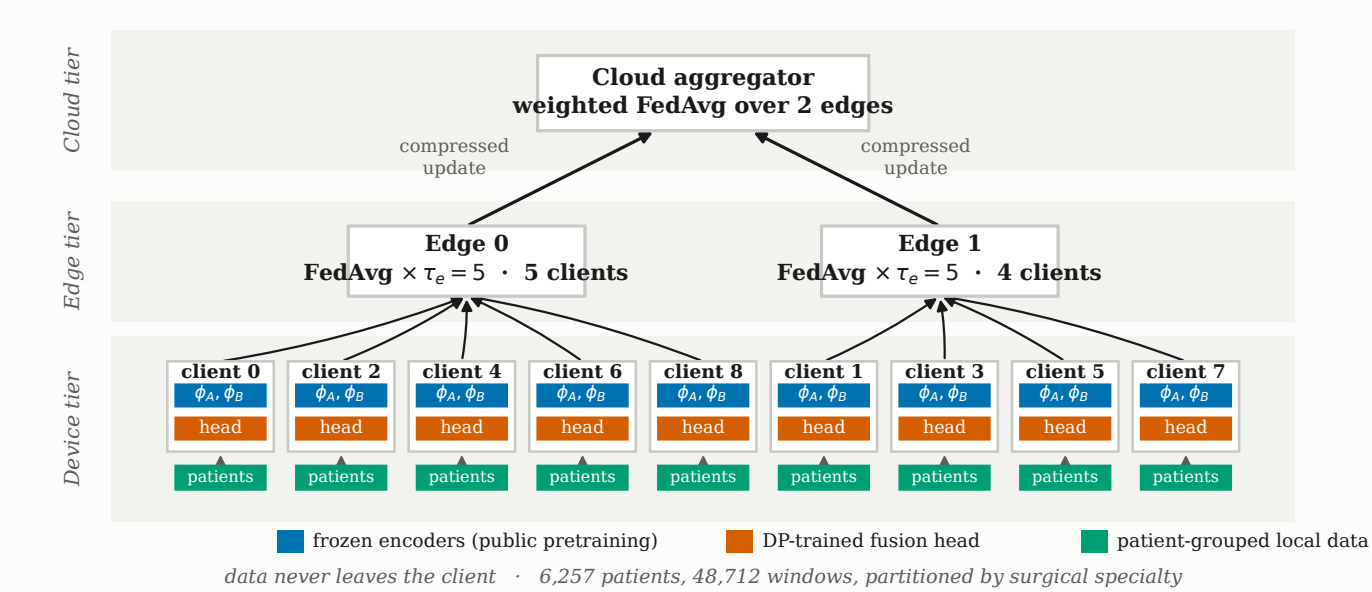


Fig. 1. Overall architecture of PP-HFM. Nine institutional clients, partitioned by surgical specialty, hold patient-grouped multimodal records that never leave the institution; each runs frozen pretrained encoders and privately trains only the fusion head. Two edge aggregators perform τ_e rounds of weighted averaging for every cloud round, and the cloud aggregator maintains and redistributes the global model. Clients are drawn grouped by edge for legibility; the indices give the true assignment.

B. Overall System Architecture

PP-HFM is a three-tier federated system built on a frozen multimodal backbone. Its guiding architectural principle is a separation of responsibilities: the encoders supply fixed representations, a compact head carries all trainable and all privatised capacity, an edge tier absorbs local rounds to relieve the wide-area link, and a cloud tier maintains the global model. The encoder design, the weighted averaging rule, and the conformal calibration procedure are adopted from prior work, and the design effort is concentrated on the unit at which privacy is enforced and on the accounting that keeps the reported budget truthful. The system is therefore best understood as a structured composition of adopted and newly specified components, each with a distinct role, rather than as a monolithic model.

The complete architecture is illustrated in Fig. 1. An institutional client holds a set of patients, each contributing a number of time-aligned multimodal records. Those records are encoded by two frozen modality-specific encoders and combined by a trainable fusion head; the head is optimised under a differentially private objective and its update, once compressed, is the only quantity that leaves the institution. Updates are averaged at the edge over the clients it serves, and after τ_e such rounds the edge models are averaged at the cloud over edge populations and redistributed. At no point does a record, a gradient of a single record, or an uncompressed update cross an institutional boundary.

At the level of module organisation the system comprises three functional groups. The first is the frozen backbone: two modality-specific encoders, pretrained on a population disjoint from every federated client and not adapted during federated training. The second is the privatised trainable component: the fusion head, together with the patient-level clipping and noise mechanism that governs its optimisation and the calibration procedure that sets the noise scale. The third comprises the communication and calibration components: the update encoding that reduces payload, the two-tier averaging that produces the global model, and the federated conformal procedure that converts point predictions into prediction sets with a coverage guarantee.

The resulting information flow is a group-then-bound structure. Within a client, records fan into the encoders and are combined by the head; gradients are then gathered *by patient* before any bound is applied, so that the quantity subsequently clipped is a patient's total contribution rather than a record's. Perturbation is applied once to the aggregate of those bounded per-patient contributions, after which the update is compressed and leaves the client. Upward, the flow is hierarchical: client to edge repeatedly, edge to cloud occasionally, with the ratio between the two set by τ_e.

The rationale follows from the two constraints the setting imposes. Grouping gradients by patient before bounding them is what makes the guarantee a statement about people, and doing so at the point of clipping rather than by a group-privacy conversion afterwards avoids the prohibitive degradation such a conversion entails. Confining private optimisation to a small head over frozen features keeps the perturbation dimensionality low, which is what makes a tight budget survivable at all; pretraining those features on a disjoint population is what keeps the resulting guarantee end-to-end rather than nominal. Interposing an edge tier reduces wide-area transmissions by the factor τ_e without altering the learning objective, and because the encoders are fixed and the aggregation rule is standard, any behaviour observed can be attributed to the privacy unit and its accounting rather than to the representation or the averaging.

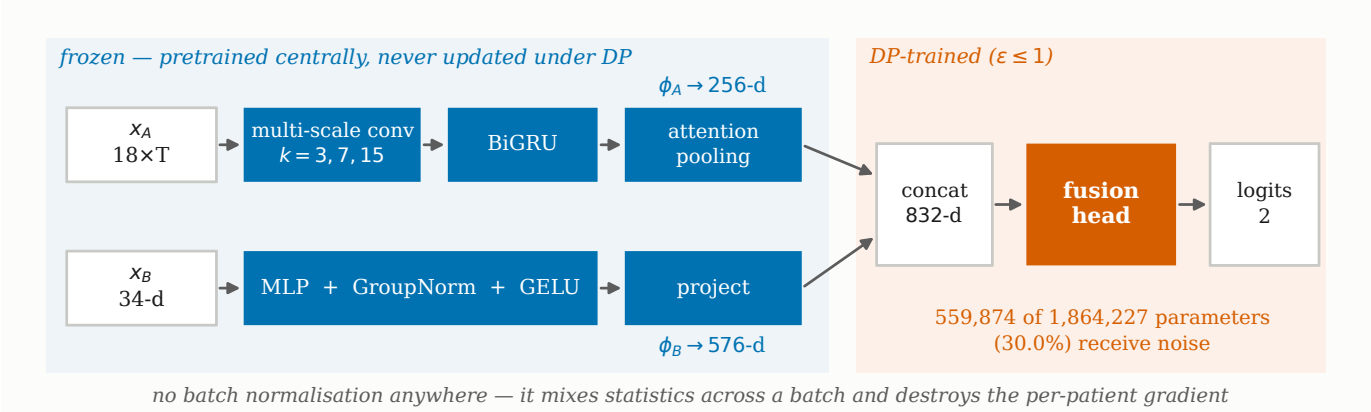


Fig. 2. Encoder–fusion backbone and the privacy boundary. Both encoders are pretrained centrally and frozen; only the fusion head is trained under differential privacy, so noise is applied to 559,874 of 1,864,227 parameters. Section III-C defines the two pretraining configurations, and Section IV-B records which produced which result. No batch normalisation appears anywhere, because it mixes statistics across a batch and would make the per-patient gradient ill-defined.

Having established the blueprint, the module organisation, and the information flow, the following subsections describe each component in detail — the frozen multimodal backbone, the patient-level privacy mechanism, the noise calibration, the hierarchical aggregation, the update encoding and conformal calibration, and the empirical audit — concluding with the formulation of the training objective and the privacy accounting.

C. Frozen Multimodal Backbone

A tight privacy budget forces a large noise scale, and the perturbation a parameter vector must absorb grows with its dimensionality, so training an entire network under such a budget degrades severely. Restricting private optimisation to a small trainable subset over frozen pretrained features is an established remedy: parameter-efficient differentially private fine-tuning is markedly more robust at small ε than end-to-end private training [11], [12]. PP-HFM adopts this design without alteration and claims no novelty for it; what the framework adds is the condition under which the resulting guarantee remains valid, and the measurement of what that condition costs.

The backbone comprises two modality-specific encoders. The temporal encoder consumes a multichannel physiological time series and applies a multi-scale convolutional stem, in which three parallel one-dimensional convolutions of different kernel widths extract fine, intermediate and coarse temporal morphology whose outputs are concatenated; a strided convolution reduces temporal resolution, two residual convolutional blocks and a bidirectional recurrent layer model longer-range dependence, and an attention-pooling layer collapses the temporal axis using learned weights. Because pooling is adaptive, the encoder accepts sequences of any length without architectural change. The static encoder consumes a fixed-length vector of preoperative clinical and laboratory variables through a two-layer perceptron with normalisation and nonlinearity. The two embeddings are concatenated and mapped to class logits by the fusion head. The backbone and head are illustrated in Fig. 2.

Two properties of the backbone are load-bearing for the privacy mechanism rather than for accuracy. The first is that its parameters are frozen throughout federated training, so the trainable and privatised capacity of the system is concentrated entirely in the fusion head, which holds 559,874 of the 1,864,227 parameters of the network; the private optimisation is therefore a near-linear probe over fixed features rather than end-to-end training. The second is that every normalisation layer is a group or layer normalisation and none is a batch normalisation. This is a requirement of the mechanism, not a tuning choice: batch normalisation mixes statistics across the examples of a batch, so a per-example — and therefore a per-patient — gradient ceases to be well defined and the accounting that follows would be unsound. The recurrent layer is likewise a per-sample-gradient-compatible implementation.

The pretraining corpus carries a condition that is easy to leave implicit and consequential to violate. If the encoders are pretrained on data belonging to the federated clients, they have already observed the protected population without a guarantee, and the budget subsequently reported for the head is not an end-to-end statement about the system. PP-HFM therefore admits two configurations of the encoders, and distinguishes them throughout.

In the *oracle* configuration the encoders are pretrained on the pooled training data of the federated clients. This configuration does *not* carry a valid end-to-end guarantee and is not proposed for deployment; it is retained because it isolates the achievable ceiling of the head-only recipe from the separate question of where the representation came from, and it is therefore the configuration under which the mechanism is characterised. In the *disjoint* configuration a set of individuals is reserved before federated training begins, disjoint at the level of the individual from every client's training, validation and test splits, and the encoders are pretrained only on that set; disjointness is verified rather than assumed. This is the configuration that carries the guarantee the paper claims, and the utility forgone by respecting it is measured rather than absorbed silently.

The distinction is stated here, in the specification of the mechanism, because a reader would otherwise reasonably assume that every reported figure was obtained under the disjoint configuration. Section IV-B records which configuration

produced which result, and Section V-A reports the cost of moving between them.

The frozen embeddings and the trainable head are the substrate on which the privacy mechanism operates. Having established what is optimised and what is not, the following subsection describes the mechanism that governs the optimisation of the head and fixes the unit at which its guarantee is stated.

D. Patient-Level Privacy Mechanism

The guarantee a clinical system must make concerns individuals, and the mechanism ordinarily used to provide it does not. Differentially private stochastic gradient descent bounds the influence of a training *record* by clipping its gradient to a fixed norm and perturbing the aggregate [5]. Where each record corresponds to a distinct person the two coincide, but in a monitoring dataset a record is a fixed-length window and a single operation yields many of them, so removing one patient perturbs as many records as that patient contributed. A record-level budget consequently bounds the wrong quantity. The relation between the two is available analytically: an (ε, δ)-differentially private mechanism is $(k\varepsilon,\ ke^{(k-1)\varepsilon}\delta)$-differentially private for groups of size k [6].

Whether the distinction arises at all is a property of the deployment, and two regimes must be separated. Where each client corresponds to a single individual — a wearable installation in which one participant carries one device — client-level, user-level and individual-level guarantees coincide, and bounding the client's contribution already bounds the person's. Where a client instead holds records belonging to many individuals — an institution holding many patients — the two units diverge for every participant, and it is this regime the mechanism below addresses. The framework applies to both; only the second exhibits the gap, and the evaluation is organised accordingly. Applied at the mean group size of this cohort that conversion turns a nominal $\varepsilon = 1$ into $\varepsilon \approx 8$ with $\delta \approx 9 \times 10^{-2}$, which is not a budget that supports a clinical claim. The gap between the two units is illustrated in Fig. 3.

The mechanism therefore enforces the guarantee at the point of clipping rather than recovering it by conversion afterwards. Its unit of sampling is the patient: at each optimisation step a client draws patients independently with a fixed rate, so that the quantity entering the accountant is a rate over the patient population rather than over the window population. For each sampled patient the gradients of all of that patient's windows are accumulated into a single per-patient gradient, and it is that sum which is bounded to the clipping norm. Bounding the sum is what makes the sensitivity patient-level: a patient can displace the aggregate by at most the clipping norm regardless of how many windows they contributed, whereas bounding each window individually would permit a displacement proportional to their number. Gaussian noise calibrated to that same norm is then added once per step to the aggregate of the bounded per-patient gradients, and the parameters of the head are updated. The internal organisation of the mechanism is shown in Fig. 4, and the procedure is stated in Algorithm 1.

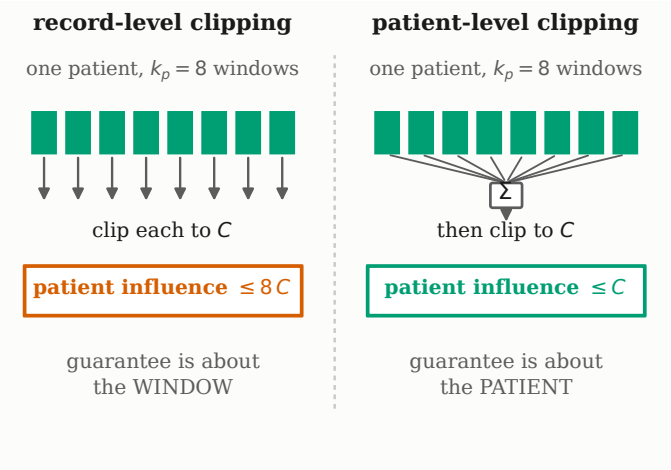


Fig. 3. Where the guarantee is decided. Clipping each window to C bounds the influence of a *window*: a patient contributing k_p windows can still displace the aggregate by k_pC. Summing a patient's windows before clipping bounds the influence of the *patient* at C, however many windows that patient contributed. k_p=8 is the cohort mean.

Algorithm 1 Patient-level private local update at client k

Require: frozen θ_{enc}; head θ_{head}; patients $\mathcal{P}_k$ with records $\{W_p\}$; clip C; noise σ; rate q from (3); step size η

for each local step **do**
  Sample $S \subseteq \mathcal{P}_k$, each p independently w.p. q
  for $p \in S$ **do**
    $g_p \leftarrow \sum_{(x,y)\in W_p} \nabla_{\theta_{\text{head}}} \ell(f_\theta(x), y)$ ▷ group by patient
    $\bar{g}_p \leftarrow g_p \cdot \min(1, C/\|g_p\|_2)$ ▷ bound the sum
  end for
  $\tilde{g} \leftarrow \frac{1}{|S|}\left(\sum_{p\in S} \bar{g}_p + \mathcal{N}(0, \sigma^2 C^2 I)\right)$
  $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta\, \tilde{g}$
  Accountant.step(σ, q)
end for
return θ_{head}

Accounting is performed with the same sampling rate that governs the sampling, so the reported budget is patient-level by construction rather than by a conversion applied after the fact, and a single accountant instance persists for the whole of a client's participation so that composition across rounds is carried rather than reset. The cost of the construction is explicit and is one of the quantities this work sets out to measure: the number of effective units per step falls from windows to patients, and a patient contributing many windows has their summed gradient bounded more aggressively than one contributing few. Whether that cost grows with the number of records per individual, and how steeply, is examined empirically rather than asserted, by holding the first effect fixed and varying the second (Section V-B3).

The mechanism fixes the unit and the form of the perturbation but not its magnitude, which must be chosen so that the budget is respected by every client rather than by a representative one. That choice is the subject of the following subsection.

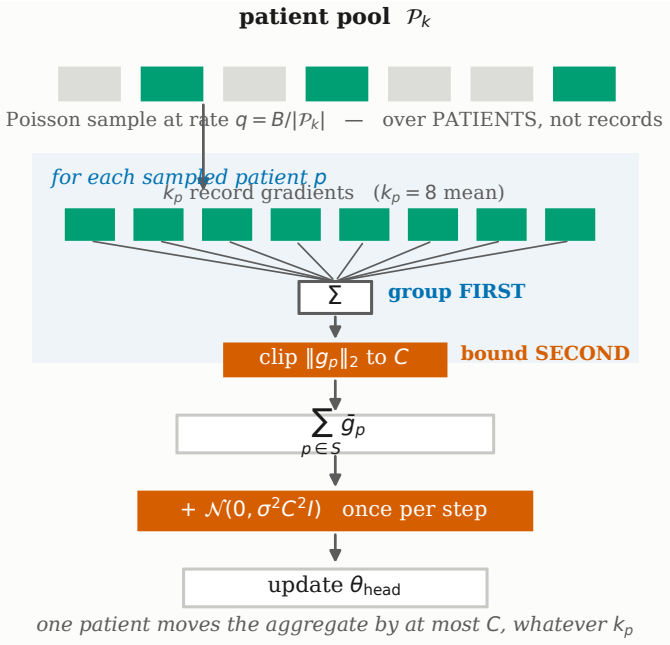


Fig. 4. Internal processing of the patient-level privacy mechanism. Patients are sampled, each sampled patient's window gradients are accumulated into one per-patient gradient, that sum is clipped to C, the clipped sums are aggregated, and a single Gaussian perturbation scaled to C is applied per step. Grouping before bounding is what fixes the unit of the guarantee.

E. Noise Calibration Across Heterogeneous Clients

The noise multiplier is the quantity that converts a clipping norm into a budget, and it is determined by the sampling rate and the number of optimisation steps a client performs. In a federated system partitioned by institution or specialty those quantities differ substantially between clients: a small client samples a large fraction of its population at each step but performs few steps, while a large client does the reverse. Neither regime dominates the other universally, and because the budget a system reports is the worst case across its clients rather than the average, a noise scale calibrated from any single client will under-provision some other client and cause the reported guarantee to be exceeded.

The calibration therefore solves for the smallest noise multiplier that satisfies the target budget simultaneously for every client, by searching over candidate values and evaluating each against all clients rather than against a representative one. Two details of that evaluation are load-bearing. The number of optimisation steps a client performs per call is obtained by rounding the ratio of its population to the batch size *upward*, because rounding downward under-counts the steps actually taken and therefore under-provisions the noise required to cover them. The population and rate for each client are read from the partition files the training procedure actually loads, rather than from any cached description of them, because a description that has drifted from the data yields a budget that is exceeded silently while every diagnostic reports values within specification.

These are stated as requirements of the method rather than as implementation notes because each corresponds to a failure that is invisible at run time: the system continues to train, the logs continue to show a budget within target, and only an independent recomputation reveals that the guarantee has been breached. A privacy mechanism whose correctness cannot be observed from its own outputs must be specified in enough detail that it can be checked from outside.

With the unit fixed, the perturbation formed and its magnitude calibrated, the locally privatised update is ready to be communicated. The following subsection describes the aggregation that converts those updates into a global model.

Algorithm 2 PP-HFM cloud round r

Require: global model $\theta^{(r-1)}$; edge map $e(\cdot)$; edge rounds τ_e

$\theta_e^{(r,0)} \leftarrow \theta^{(r-1)}$ for all edges e
for $s = 1$ **to** τ_e **do**
  for each client k **in parallel do**
    $\theta_k^{(r,s)} \leftarrow$ Algorithm 1 from $\theta_{e(k)}^{(r,s-1)}$
    Transmit $Q\big(\text{top-}\kappa(\theta_k^{(r,s)} - \theta_{e(k)}^{(r,s-1)})\big)$
  end for
  Each edge aggregates by (8)
end for
Cloud aggregates by (9) and broadcasts $\theta^{(r)}$
if final round **then**
  Clients release perturbed quantiles of (12); aggregate to $\hat{q}$
end if
return $\theta^{(r)}$, $\hat{q}$

F. Two-Tier Hierarchical Aggregation

A flat federated system transmits every client update to the cloud in every round, which makes the wide-area link the binding constraint on how often the model can be updated. Interposing an aggregation tier between clients and cloud relieves that constraint [2], [3]: several rounds of local optimisation and edge averaging are performed for each round of cloud averaging, so the expensive link is exercised once per group of local rounds rather than once per round. PP-HFM adopts this organisation, with the number of edge rounds per cloud round denoted τ_e, and claims no novelty for the averaging rule itself.

Aggregation proceeds in two stages, both by averaging weighted by the number of training records a participant holds. Within a cloud round, each client initialises from the model held by its edge, performs one locally private pass, and transmits its encoded update; the edge forms the weighted average over the clients it serves and returns it, and this repeats τ_e times. The edges then report to the cloud, which forms the weighted average over edge populations and broadcasts the result downward as the global model for the next round. Parameters that are not floating-point are carried through unchanged rather than averaged. The full procedure across both tiers is stated in Algorithm 2.

The parameter τ_e governs the trade the hierarchy makes. Setting it to one collapses the edge tier into the cloud tier, recovering conventional single-tier federated averaging, which is the configuration used to isolate the contribution of

the hierarchy; increasing it reduces wide-area transmissions proportionally while allowing edge models to diverge further between synchronisations. Because the privacy mechanism operates entirely within a client and the aggregation consumes only already-privatised updates, τ_e affects communication and convergence but does not enter the guarantee except through the number of optimisation steps it implies, which the calibration of the preceding subsection already accounts for.

The updates the two tiers exchange are not transmitted in their raw form, and the model they produce emits point predictions that a clinical setting requires to be qualified. The encoding of the updates and the calibration of the predictions are described together in the following subsection.

G. Update Encoding and Conformal Calibration

Two components complete the system, both adopted from established practice and neither claimed as a contribution. The first reduces the payload the hierarchy transmits. A client update is sparsified by retaining only its largest-magnitude coordinates and the retained values are quantised uniformly to a reduced bit width; the residual discarded by sparsification is retained locally and added to the update of the following round, so that coordinates suppressed in one round are eventually transmitted rather than lost. The encoding is applied to an update that has already been perturbed by the privacy mechanism, and is therefore a post-processing operation on a private quantity; by the post-processing invariance of differential privacy it does not affect the guarantee, which is why it can be tuned for bandwidth without privacy consequence.

The second qualifies the predictions. A point prediction is insufficient for clinical use, and conformal prediction supplies prediction sets with a distribution-free coverage guarantee that requires no retraining [23], [24]. Each client computes nonconformity scores on a held-out calibration split and releases a quantile of them; the edges aggregate client quantiles, the cloud aggregates edge quantiles, and the resulting global threshold defines the prediction set. The released quantile is, however, a statistic of protected data, and its release is a disclosure in exactly the sense the framework is designed to bound. It is therefore perturbed before release and, more importantly, is *budgeted*: the calibration term is allocated a share of the total budget in advance and the composed total is what the system reports.

That accounting decision deserves emphasis because the alternative fails silently. If calibration is triggered once per edge round rather than once per training run, its cost composes across every edge round; a system that reports only the training term then understates the budget it has spent by a multiplicative factor while every logged value remains within its stated target. Splitting the budget in advance and reporting the composed total removes that failure mode by construction rather than by vigilance.

The framework as described provides a guarantee whose parameters are known by construction. Whether that guarantee corresponds to what an adversary can actually achieve is a separate question, and it is the subject of the following subsection.

H. Deployment Path and Inference Under a Latency Budget

The training procedure described so far produces a model; the clinical value of that model depends on the delay between a measurement becoming available and a prediction being returned, which is a property of the deployment path rather than of the optimisation. That path is therefore specified as part of the system. The trained network is exported to a portable computation graph and quantised to integer precision, a transformation that reduces both memory footprint and arithmetic cost on commodity hardware without retraining [26] and which is applied after federated training has completed, so that it operates on an already-released model and has no bearing on the privacy guarantee. Inference is executed on the edge tier rather than in the cloud, so that the wide-area link is not on the critical path of a prediction.

The delay is decomposed into four stages, each measured separately rather than inferred from a single end-to-end figure, because a budget that is satisfied overall may still be dominated by a stage that a deployment could remove. Preprocessing constructs and normalises the input tensors from the sensor stream; the forward pass evaluates the exported graph; the conformal stage constructs the prediction set from the resulting logits using the global threshold obtained during calibration; and a final stage converts the outcome into the form the consuming system expects. Reporting the stages separately also locates the cost of the conformal guarantee explicitly, which matters because a coverage guarantee that doubled the response time would be a poor trade in a setting where the response time is the binding constraint.

Latency is characterised by its distribution rather than by its mean. The system is required to satisfy its budget at the ninety-fifth percentile, since a mean conceals precisely the tail that determines whether a deployment meets its obligations under load, and a clinical service level is stated as a percentile rather than an average. The measurement configuration is fixed in advance rather than chosen after inspection of the results: the integer- quantised graph is executed on a general-purpose CPU provider, representing an edge node without an accelerator, for one thousand timed executions preceded by one hundred discarded warm-up executions, and each of the four stages is timed separately within every execution. Reporting against a CPU provider is deliberate; an edge deployment cannot be assumed to have a GPU, and a latency figure obtained on one would not describe the setting the system targets.

I. Network and Energy Instrumentation

Three of the system's four design objectives concern quantities that a training script cannot observe: the volume of traffic the hierarchy carries over the wide-area link, the reliability with which updates arrive, and the energy each tier expends in producing a round. These are properties of a deployed topology rather than of an optimisation, and are therefore measured on instrumented models of that topology rather than estimated from parameter counts.

The network is modelled as the three tiers of Section III-B, with the device-to-edge segment carried over a wireless local-area link and the edge-to-cloud segment over a point-to-point

TABLE I
DESIGN OBJECTIVES AND THEIR PRE-REGISTERED TARGETS. THE PROBLEM SPECIFICATION FIXES A NUMERIC THRESHOLD FOR COMMUNICATION REDUCTION BUT ASKS ONLY THAT THE LATENCY BENEFIT OF THE HIERARCHY BE QUANTIFIED, SO O1B IS STATED AS A DIRECTION RATHER THAN A PERCENTAGE. MEASURED VALUES ARE REPORTED IN SECTION V.

	Quantity	Target
O1	Uplink traffic reduction vs. centralised	$\geq 50\%$
O1b	Training-round latency vs. centralised	improvement
O2a	Privacy budget, composed total	$\varepsilon \leq 1$
O2b	Accuracy loss under the budget	≤ 2 pp
O3a	Inference latency, 95th percentile	< 100 ms
O3b	Successful model updates	$> 99\%$
O4	Energy saving per round vs. centralised	$\geq 20\%$

backhaul, so that the asymmetry the hierarchy exploits — frequent local exchange over a cheap link, infrequent global exchange over an expensive one — is represented rather than assumed. Each round is simulated in both directions, the downlink broadcast of the global model and the uplink transmission of client updates, and the volume, delivery reliability and end-to-end delay are recorded per round for the proposed configuration and for the flat and centralised alternatives against which it is compared. Recording the delay per round is what allows the latency benefit of the hierarchy to be quantified separately from the latency of serving a prediction: the former is a property of how often the wide-area link is traversed during training, the latter of the deployment path of Section III-H, and a design may improve one without affecting the other. Because the payload is the compressed update of Eqs. (10)–(11), the traffic measured is that of the system as specified rather than of an uncompressed idealisation.

Energy is modelled per tier from the power characteristics of the participating hardware and the completion time of the work assigned to it, with the device tier representing constrained embedded nodes, the edge tier representing server-class accelerated hardware, and the cloud tier representing the aggregation host. Attributing energy by tier is what makes the comparison informative: a hierarchical design does not reduce the total work performed so much as move it, and a figure reported only in aggregate would conceal whether a saving at one tier had been purchased by an increase at another.

The targets against which these quantities are assessed are fixed in advance and are stated in Table I; the measured values are reported in Section V. Stating the targets before the measurement, and in a single place, is what prevents a threshold from being adjusted after the fact to accommodate a result.

J. Empirical Auditing of the Guarantee

A differential privacy guarantee is an upper bound on the influence of an individual; it does not establish what an adversary attains in practice, and in particular it does not establish that a system providing a weaker guarantee is exploitable. Membership inference supplies the complementary measurement [4]. PP-HFM is audited by mounting membership inference at the level of the individual: members are the

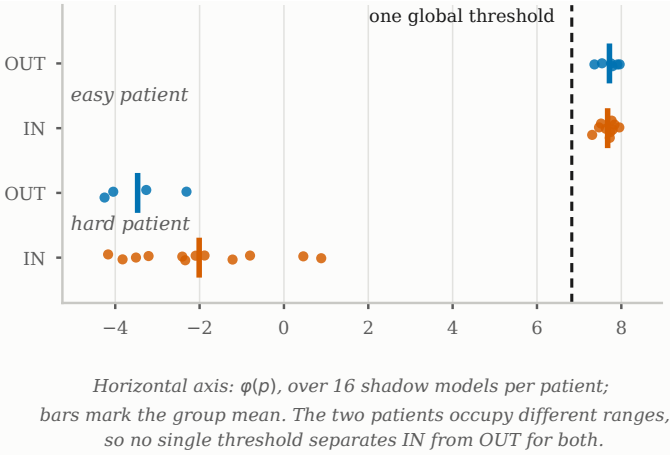


Fig. 5. Why per-patient calibration is necessary. Under a single global threshold an easy non-member can outscore a hard member, so membership signal is buried in per-patient difficulty. Fitting each patient's own in- and out-distributions from shadow models removes that confound and scores membership as a likelihood ratio.

patients held in the clients' training splits, non-members are patients held out entirely, and both groups are scored by the same released model through the same procedure, so that any separation between them reflects membership rather than a difference in how they were handled.

The audit uses two attacks of deliberately different strength. The first compares each patient's mean per-record loss against a global threshold [17]. Aggregating to the patient before deciding is the essential step: a per-record attack measures record membership, which is precisely the quantity a record-level guarantee does protect, whereas averaging a patient's records is what converts weak per-record signal into per-person signal. The second is a likelihood-ratio attack [13], adapted so that the unit of membership is the patient. Shadow models are trained on independently drawn halves of the patient population, with all of a patient's records moving together and each shadow trained under the same recipe as the model it calibrates; for each patient the distribution of a logit-scaled confidence statistic is estimated separately over the shadows in which that patient was a member and those in which it was not, and membership is scored by the ratio of the two. The distinction matters because a global threshold is confounded by per-patient difficulty, so the strength it reports is a lower bound on leakage rather than an estimate of it [18]; the per-patient calibration removes that confound, as illustrated in Fig. 5.

Results are reported by area under the receiver operating characteristic curve and additionally by true-positive rate at low false-positive rate, because an average-case summary conceals the tail: an attack near chance on average may still identify a small number of individuals with high confidence, and for a disclosure claim those identifications are the harm. Where the number of non-members is insufficient to resolve a requested false-positive rate the value is reported as undefined rather than interpolated.

The audit carries one methodological requirement without which its verdict would be uninterpretable. An attack that finds

TABLE II
NOTATION.

Symbol	Meaning
K, E	number of clients, of edge aggregators
$\mathcal{P}_k$, W_p	patients at client k; records of patient p
$k_p = \lvert W_p \rvert$	records contributed by patient p
R, τ_e	cloud rounds; edge rounds per cloud round
B	patients sampled per optimisation step
C, σ	clipping norm; noise multiplier
q	Poisson sampling rate over patients
(ε, δ)	privacy budget
α	conformal miscoverage level
θ_{enc}, θ_{head}	encoder, fusion-head parameters

nothing is indistinguishable from an attack that is incapable of finding anything, so the same procedure is additionally applied to a configuration known to memorise — non-private training of the same head over the same frozen encoders — and is required to register clearly elevated leakage there before any negative result elsewhere is admitted. This positive control is a precondition of the audit rather than a supplementary experiment, and it is reported alongside every result it qualifies.

Having specified the mechanism, its calibration, the aggregation that combines its outputs, the components that encode and qualify them, and the audit that tests the guarantee they provide, the following subsection states the training objective and the accounting formally.

K. Training Objective and Privacy Accounting

Let K clients be indexed by k, with client k holding a local dataset $\mathcal{D}_k$ of n_k records and $n = \sum_k n_k$. Clients are assigned to edges by a fixed map $e(\cdot)$. The records of client k are partitioned by individual into a set of patients $\mathcal{P}_k$, and W_p denotes the records contributed by patient p, with $k_p = \lvert W_p \rvert$; the partition is constructed so that W_p never spans two clients or two evaluation splits. Model parameters are written $\theta = (\theta_{\mathrm{enc}}, \theta_{\mathrm{head}})$, and ℓ is the class-weighted cross-entropy. The notation is summarised in Table II.

The learning objective is the standard weighted empirical risk over clients,

$$\begin{aligned} \min_{\theta}\ F(\theta) &= \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \\ F_k(\theta) &= \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell\big(f_\theta(x), y\big), \end{aligned} \tag{1}$$

minimised only over θ_{head}, since θ_{enc} is pretrained on a disjoint public population and held fixed,

$$\theta_{\mathrm{enc}} \text{ fixed}, \qquad \theta_{\mathrm{head}} \leftarrow \text{privately optimised}. \tag{2}$$

At each optimisation step client k draws a subset $S \subseteq \mathcal{P}_k$ by including each patient independently with probability

$$q = \frac{B}{\lvert\mathcal{P}_k\rvert}, \tag{3}$$

a rate defined over patients rather than over records, which is the substitution that makes the accounted quantity patient-level. For each sampled patient the record gradients are accumulated and the sum is bounded,

$$\begin{aligned} g_p &= \sum_{(x,y)\in W_p} \nabla_{\theta_{\mathrm{head}}}\, \ell\big(f_\theta(x), y\big), \\ \bar{g}_p &= g_p \cdot \min\!\left(1, \frac{C}{\lVert g_p \rVert_2}\right). \end{aligned} \tag{4}$$

Equation (4) is the step at which the unit of the guarantee is determined: because the bound is applied to the sum, the influence of patient p is at most C irrespective of k_p, whereas bounding each record separately would admit an influence of $k_p C$. A single perturbation scaled to the same norm is then applied to the aggregate,

$$\tilde{g} = \frac{1}{\lvert S\rvert}\left(\sum_{p\in S} \bar{g}_p + \mathcal{N}\big(0, \sigma^2 C^2 I\big)\right), \tag{5}$$

and the head is updated by $\theta_{\mathrm{head}} \leftarrow \theta_{\mathrm{head}} - \eta\,\tilde{g}$.

The noise multiplier is the smallest value satisfying the budget for every client simultaneously,

$$\sigma^\star = \min\Big\{\sigma :\ \max_k\ \varepsilon_k\big(\sigma,\, q_k,\, T_k\big) \le \varepsilon_{\mathrm{train}}\Big\}, \tag{6}$$

where ε_k is obtained from a Rényi accountant [16] and the step count is

$$T_k = R\,\tau_e \left\lceil \frac{n_k}{B} \right\rceil. \tag{7}$$

The maximum in Eq. (6) is taken over all clients rather than a representative one, and the ceiling in Eq. (7) rounds upward; either substituted for its convenient alternative yields a budget that is exceeded without any observable symptom.

Aggregation applies the weighted average at each tier. Writing $\theta_k^{(r,s)}$ for the model of client k after edge round s of cloud round r,

$$\theta_e^{(r,s)} = \sum_{k:\, e(k)=e} \frac{n_k}{\sum_{j:e(j)=e} n_j}\, \theta_k^{(r,s)}, \tag{8}$$

$$\theta^{(r)} = \sum_{e=1}^{E} \frac{N_e}{\sum_{e'} N_{e'}}\, \theta_e^{(r,\tau_e)}, \qquad N_e = \sum_{k:e(k)=e} n_k. \tag{9}$$

Updates are encoded before transmission. Sparsification retains the κd largest-magnitude coordinates of an update Δ of dimension d,

$$[\Delta^{\mathrm{sp}}]_i = \begin{cases} [\Delta]_i, & i \in \arg\operatorname{top}_{\kappa d} \lvert\Delta\rvert \\ 0, & \text{otherwise,} \end{cases} \tag{10}$$

and the retained values are quantised uniformly to b bits with step $s = (v_{\max} - v_{\min})/(2^b - 1)$,

$$Q(v) = v_{\min} + \operatorname{round}\!\Big(\frac{v - v_{\min}}{s}\Big)\, s. \tag{11}$$

The residual $\Delta - \Delta^{\mathrm{sp}}$ is accumulated locally and added to the next round's update. Both operations act on the already-perturbed quantity of Eq. (5) and are therefore post-processing.

Conformal calibration uses the least-ambiguous-classifier score on a held-out split,

$$r_i = 1 - \hat{p}_\theta(y_i \mid x_i)\,, \tag{12}$$

whose perturbed quantile at level $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / n_{\mathrm{cal}}$ is released, aggregated across tiers to a global threshold $\hat{q}$, and used to form the prediction set $\mathcal{C}(x) = \{y : 1-\hat{p}_\theta(y \mid x) \le \hat{q}\}$. The budget spent by a client is the composition of the training and calibration terms,

$$\varepsilon_{\mathrm{total}} = \varepsilon_{\mathrm{train}} + \varepsilon_{\mathrm{conf}}, \quad \varepsilon_{\mathrm{train}} = 0.8\,\varepsilon, \quad \varepsilon_{\mathrm{conf}} = 0.2\,\varepsilon, \tag{13}$$

and it is $\varepsilon_{\mathrm{total}}$, not $\varepsilon_{\mathrm{train}}$, that the system reports.

Finally, the audit statistic of Section III-J is the logit-scaled confidence of a patient, averaged over that patient's records,

$$\varphi(p) = \frac{1}{|W_p|} \sum_{(x,y)\in W_p} \Big(z_y(x) - \log \sum_{j \neq y} e^{z_j(x)} \Big), \tag{14}$$

which is the numerically stable form of $\log\big(\hat{p}_y/(1-\hat{p}_y)\big)$, and membership is scored by the ratio of the fitted in- and out-densities,

$$\Lambda(p) = \log \frac{\mathcal{N}\big(\varphi(p);\, \mu_{\mathrm{in}}(p), \sigma^2_{\mathrm{in}}(p)\big)}{\mathcal{N}\big(\varphi(p);\, \mu_{\mathrm{out}}(p), \sigma^2_{\mathrm{out}}(p)\big)}. \tag{15}$$

The objective in Eq. (1) is minimised over the fusion head alone, under the mechanism of Eqs. (3)–(5) with the noise scale fixed by Eq. (6), while the encoders remain frozen. Optimisation uses stochastic gradient descent on the perturbed gradient, with the round at which the reported model is taken selected on validation performance and the composed budget of Eq. (13) reported in place of the training term alone. Having specified the framework and its accounting, the following section describes the datasets, the implementation, and the evaluation protocol.

IV. Experimental Setup

A. Datasets

The proposed framework is evaluated on two publicly available multimodal corpora: VitalDB [21], a clinical monitoring collection of non-cardiac surgical cases, and WEAR [22], an outdoor activity collection combining body-worn inertial sensing with egocentric video. Both were selected on a property that is decisive for a claim about fusion: in each, the two modalities originate from the same individual over the same interval and are associated by the recording apparatus rather than by a construction rule. Where modalities are paired by a rule, an apparent benefit of fusion may reflect the regularity of that rule rather than complementary information, and no claim about fusion can be sustained; neither corpus used here requires such a defence. Together they span the two deployment regimes that matter for the privacy mechanism — a client holding many individuals, and a client holding one.

In VitalDB, modality A is the intraoperative monitoring record represented as eighteen channels, comprising nine numeric physiological trends and nine binary indicators of whether each trend was recorded, and modality B is a thirty-four-dimensional vector of preoperative clinical and laboratory variables. Every channel of a case originates from the same patient during the same operation, time-aligned at capture across the waveform digitiser, the patient monitor and the anaesthesia machine. The prediction target is postoperative admission to intensive care, positive in $19.65\,\%$ of windows. In WEAR, modality A is twelve channels of inertial sensing and modality B is a 2048-dimensional two-stream video feature computed over the same interval and synchronised in hardware, with activity recognition over eighteen classes as the target.

Each corpus is partitioned into disjoint training, validation and test subsets, and these partitions are held fixed throughout all experiments so that no record contributes to more than one subset and no arm of any comparison sees a different division of the data. For VitalDB the partition comprises $34{,}044$ training, $7{,}308$ validation and $7{,}360$ test windows drawn from $6{,}257$ operations, distributed across nine clients by surgical specialty and assigned to two edge aggregators; an individual contributes 7.8 windows on average. For WEAR the partition comprises $26{,}424$ training, $5{,}395$ validation and $5{,}969$ test windows across twenty-two clients, one per participant, likewise assigned to two edges. Partitioning by specialty rather than at random produces heterogeneity of the kind that arises in deployment: specialties differ in case mix, in outcome prevalence and in which channels are routinely recorded, as Fig. 6 shows. The validation subset is used exclusively for model selection and the test subset is reserved for final evaluation, never consulted during training or checkpoint selection.

Four controls govern the preparation of the clinical corpus, each addressing a way in which a reported figure could be inflated with no error visible during training. First, splitting is performed by case, so that every window of one operation falls on the same side of every boundary; windows minutes apart from one patient are highly correlated, and a window-level split would place near duplicates of a training example into the test set. Second, modality B is restricted to preoperative variables, every intraoperative column being excluded, because interventions such as transfusion or vasopressor administration are given *because* a patient is deteriorating and therefore encode much of the outcome. Third, missingness is retained as signal rather than imputed away: gaps are filled within a case and paired with an explicit binary indicator rather than silently zero-filled, since an arterial line is placed only in patients judged at risk and its absence is itself evidence about acuity. Fourth, normalisation statistics are computed with missing-value-aware estimators over training cases only, because a single all-missing channel under a naive estimator yields an undefined statistic that propagates to every window of every case, destroying that channel while training continues to report an accuracy. A further set of individuals is reserved before federated training and used only to pretrain the encoders, disjoint at the level of the individual from every client's data, which is what makes the reported budget an end-to-end statement rather than one conditioned on encoders that had already observed the protected population.

B. Implementation Details

All arms are trained under a single, consistent configuration so that differences in behaviour can be attributed to the mechanism under test rather than to the training regime. Each client performs one local pass per federated call; five edge rounds

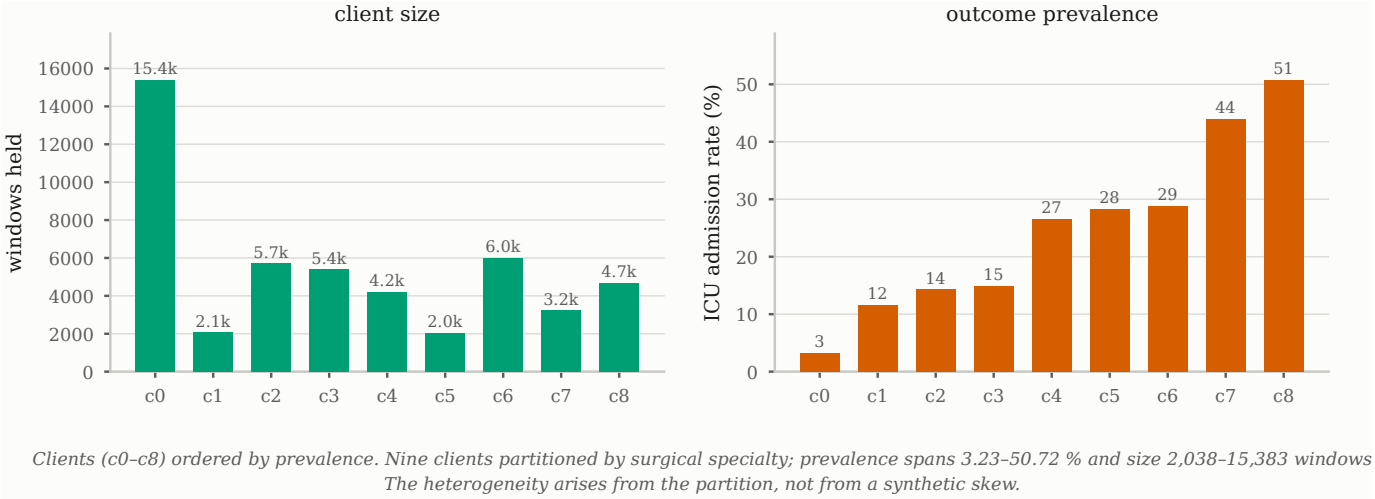


Fig. 6. Heterogeneity of the clinical partition. Partitioning by surgical specialty rather than at random yields clients that differ by a factor of seven in size and by more than fifteen in outcome prevalence, from $3.23\,\%$ to $50.72\,\%$. The skew is a property of how care is organised, not an artefact imposed for the experiment.

are executed for every cloud round, and training proceeds for ten cloud rounds. Optimisation is stochastic gradient descent on the privatised gradient, with the per-individual clipping norm set to unity and the noise multiplier obtained by the calibration of (6) against every client simultaneously. Updates are sparsified to the largest twenty per cent of coordinates and quantised to eight bits before transmission, with the residual retained locally as error feedback. Conformal calibration is performed once per run at a miscoverage level of 0.1, and its cost is composed into the reported budget according to (13).

The encoders are pretrained centrally and then frozen, so that the privatised parameter vector is the fusion head alone: $559{,}874$ of the $1{,}864{,}227$ parameters of the network, or $30.0\,\%$.

Which encoder configuration produced which result is recorded explicitly, because the two are not interchangeable. Unless stated otherwise, every result in this paper — the system-level objectives, the privacy–utility comparisons and the membership-inference audit — was obtained under the *oracle* configuration of Section III-C, in which the encoders were pretrained on the pooled client training data. That configuration does not carry a valid end-to-end guarantee; it is used because it isolates the behaviour of the privatised head from the provenance of the representation beneath it, and because holding it fixed across every arm keeps the comparisons between arms clean. The *disjoint* configuration, which does carry the guarantee, is evaluated separately, and the difference between the two is reported in Section V-A as the cost of a valid guarantee. Readers should treat the absolute utilities reported here as an upper bound on what the deployable configuration attains, and the differences between arms — which are what the claims rest on — as unaffected, since every arm shares the same encoders. Every normalisation layer in the network is a group or layer normalisation and none is a batch normalisation, for the reason given in Section III-C: batch statistics couple the examples of a batch and would make the per-individual gradient ill-defined.

Models are implemented in PyTorch 2.5, with per-sample gradient machinery and Rényi accounting provided by Opacus 1.5. Deployment artefacts are exported to ONNX and quantised to eight-bit integer precision, and inference is measured under ONNX Runtime on a general-purpose CPU provider, since an edge node cannot be assumed to carry an accelerator and a latency figure obtained on one would not describe the target setting. Network behaviour is simulated in ns-3 3.46, with the device-to-edge segment carried over a wireless local-area link and the edge-to-cloud segment over a point-to-point backhaul; energy is modelled in CloudSim Plus and attributed separately to the device, edge and cloud tiers. Training was performed on a single workstation with one NVIDIA RTX A2000 accelerator.

C. Evaluation Metrics

Classification quality is assessed with two complementary measures. The area under the receiver operating characteristic curve summarises ranking performance across all decision thresholds. It is reported alongside average precision, which summarises the precision–recall trade-off,

$$\mathrm{AP} = \sum_{n} \left(R_n - R_{n-1}\right) P_n, \tag{16}$$

where P_n and R_n are the precision and recall at the nth threshold. Average precision is reported because the clinical task is substantially imbalanced at $19.65\,\%$ positive: accuracy is nearly uninformative, since a constant negative prediction attains four-fifths of it, and the area under the ROC curve is optimistic under imbalance. Average precision is the quantity that responds when the minority class is handled well.

Metrics are additionally aggregated to the individual. The model scores windows, but the clinical question concerns an operation, and the windows of one case are highly correlated,

so a window-level metric overstates the effective sample size. The predictions of a case are therefore averaged,

$$\bar{p}_c = \frac{1}{|W_c|} \sum_{w \in W_c} p_w, \tag{17}$$

yielding one prediction per individual, which is both the clinically meaningful unit and the statistically honest one. Case-level metrics are computed from $\bar{p}_c$ by the same expressions applied to p_w.

Comparisons between arms are paired by seed rather than reported as independent means, since the arms share their partition, their initialisation schedule and their calibration data. For S paired seeds with per-seed difference d_s, the reported interval is

$$\bar{d} \pm t_{S-1,\,0.975}\, \frac{s_d}{\sqrt{S}}, \tag{18}$$

with s_d the sample standard deviation of the differences. Student's t is used rather than a normal approximation because the seed counts are small enough for the difference to be material. Seed counts are fixed in advance of each comparison and are not extended after inspecting an interval.

Communication, latency, reliability and energy are reported as defined in Section III-I, with every network quantity measured on a round in which the payload traversed the simulated topology.

D. Experimental Protocol

The proposed system is compared against differentially private federated averaging in three configurations, each isolating one design decision. The first trains the full parameter vector under the privacy budget, which is the conventional application of DP-SGD to a federated model. The second restricts private optimisation to the fusion head over randomly initialised frozen encoders, isolating the effect of reducing the privatised dimensionality alone. The third restricts private optimisation to the fusion head over *pretrained* frozen encoders, and is included specifically as a confound control: without it the benefit of parameter-efficient private training cannot be distinguished from the benefit of the pretrained representation on which it happens to sit. A non-private reference trained under an otherwise identical recipe supplies the reference point for the accuracy cost of the budget. All arms share the partition, the schedule, the compression settings and the evaluation procedure, differing only in the factor under test.

A single canonical protocol is applied to every arm. The checkpoint attaining the highest validation performance is retained and carried forward; no test-subset information influences that selection. Each retained checkpoint is evaluated once on the held-out test subset, and every metric is computed from the same predictions under the same aggregation. Applying one procedure to every arm ensures that reported differences reflect the mechanisms rather than differences in measurement.

The privacy guarantee is audited rather than reported from its parameters alone. Membership inference is mounted at the level of the individual under the two attacks of Section III-J, and every attack is additionally applied to a configuration known to memorise — non-private training of the same head over the same frozen encoders — which serves as a positive control. The control is a precondition of the audit rather than a supplementary experiment: an attack that finds nothing is indistinguishable from an attack incapable of finding anything, so no negative result is admitted unless the same procedure registers clearly elevated leakage on the control.

A further experiment separates the two mechanisms through which the choice of unit imposes a cost. Moving the guarantee from the record to the individual both reduces the number of units available to be sampled at each step and bounds the summed contribution of an individual, and the aggregate comparison cannot attribute the cost between them. The number of units drawn per step is therefore held equal in the two arms, which removes the first mechanism by construction and leaves only the clipping geometry, and the number of records retained per patient in training, k, is varied over $\{1, 2, 4, 8\}$ at three seeds under $\varepsilon = 1$. Subsampling is applied to the training partition alone; validation and test are left whole, so every point is evaluated against the same held-out data. Reducing k also shrinks the training set, so absolute utility falls with k and the levels are not comparable across k; the quantity of interest is the paired difference between the arms at each k, where both arms see the same data, the same partition and the same seed. The design is falsifiable at $k = 1$, where an individual contributing one record is a record and the two arms must therefore agree.

The following section reports these experiments: first the system-level objectives on both deployments, then the utility and privacy comparisons against the baselines, and finally the membership-inference audit.

V. Results

A. System-Level Objectives

The framework is evaluated on two deployments that differ in the property most relevant to it. In the clinical deployment, nine institutional clients hold the records of $6{,}257$ patients, so a client contains many individuals and each individual contributes 7.8 records on average. In the wearable deployment, each of twenty-two clients corresponds to a single participant contributing some $1{,}750$ records. The two therefore exercise the same framework under opposite groupings, and the second serves as a check that the system-level behaviour is not particular to one topology or payload size.

Table III reports the measured value of each design objective of Table I on both. Every target is attained on the clinical deployment; on the wearable deployment every target is attained except the accuracy bound, which is inconclusive for the reason given below. Two of the measurements warrant comment before the individual objectives are discussed, because both concern how the numbers were obtained rather than what they are.

First, the network quantities are measured rather than modelled. An earlier configuration of the simulator evaluated the packet-level topology on one round in ten and composed the remaining rounds analytically, and the two do not agree: the analytical model placed the communication term of the

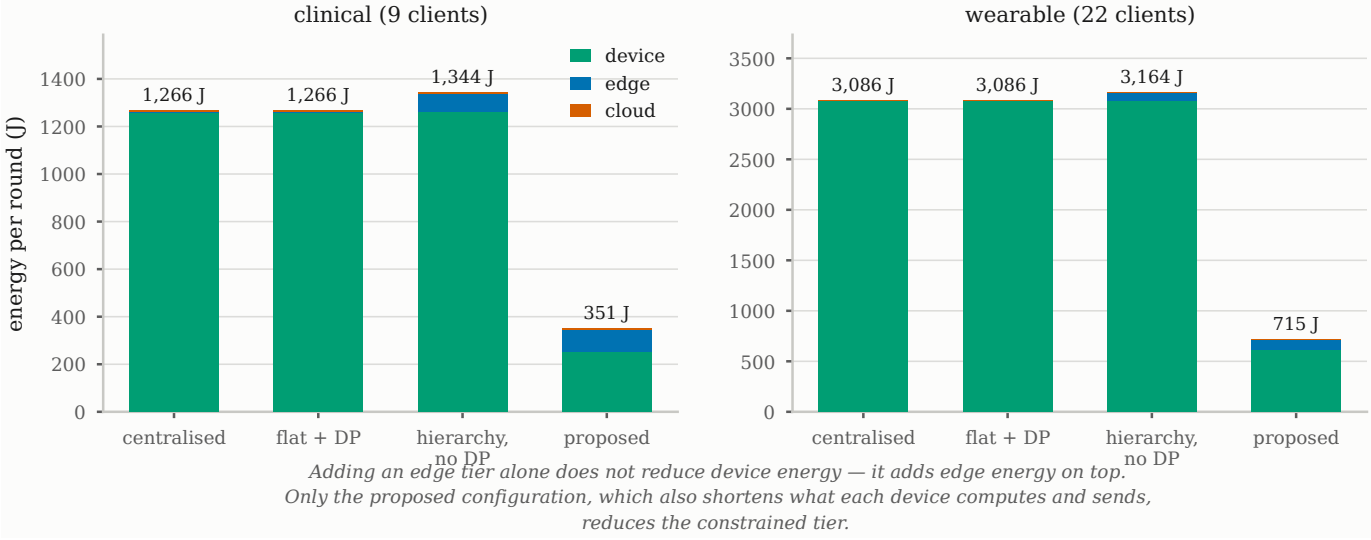


Fig. 7. Energy per training round, attributed by tier, on both deployments. The hierarchy alone relocates work without reducing it — device energy is unchanged and edge energy is added — and only the proposed configuration reduces the constrained tier.

TABLE III
MEASURED OUTCOMES AGAINST THE PRE-REGISTERED TARGETS OF TABLE I. NETWORK QUANTITIES ARE MEASURED AT PACKET LEVEL ON EVERY SIMULATED ROUND OF BOTH DEPLOYMENTS. THE ACCURACY ROW IS A PAIRED COMPARISON AGAINST A NON-PRIVATE REFERENCE TRAINED UNDER THE IDENTICAL RECIPE; † MARKS AN INTERVAL THAT SPANS ITS THRESHOLD AND IS THEREFORE INCONCLUSIVE AT THE PRE-REGISTERED SAMPLE SIZE.

	Quantity	Clinical	Wearable	Target
O1	Uplink reduction	90.8%	96.1%	$\geq 50\%$
O1b	Round latency	16.9%	21.7%	improvement
O2a	Budget (composed)	0.970	0.966	$\varepsilon \leq 1$
O2b	Accuracy change	-0.92 pp	$1.97^{\dagger}$ pp	≤ 2 pp
O3a	Latency (P95)	4.58 ms	1.22 ms	< 100 ms
O3b	Update success	100.00%	99.94%	$> 99\%$
O4	Energy saving	72.3%	76.8%	$\geq 20\%$

TABLE IV
INFERENCE LATENCY DECOMPOSITION, INTEGER-QUANTISED GRAPH ON A CPU PROVIDER, 1000 TIMED EXECUTIONS AFTER 100 DISCARDED WARM-UP EXECUTIONS.

Stage	Mean (ms)	P95 (ms)
Preprocessing	0.140	0.147
Forward pass	4.221	4.395
Conformal set construction	0.030	0.037
Post-processing	0.011	0.014
Total	4.401	4.582

centralised baseline at 256 ms where the simulated network delivers 23.6 ms, which inflated the apparent latency benefit of the hierarchy roughly fourfold. Every network figure reported here is therefore drawn from rounds in which the payload traversed the simulated topology, with no analytically composed rows contributing to any mean.

Second, the reported privacy budget is the composed total. Training accounts for $\varepsilon = 0.7705$ and the release of the conformal calibration quantile for a further 0.2000, giving 0.970 against a budget of unity. Had only the training term been reported — which is the quantity a naive accounting produces — the system would have appeared to spend 23 % less than it did.

1) Communication and latency: Two-tier aggregation reduces uplink traffic by 90.8 % relative to centralised training, against a required 50 %. The measurement is taken from the compressor as deployed and includes the index array that top-κ sparsification must transmit alongside the retained values, which an analytical payload estimate omits and which is not negligible at high sparsity.

The hierarchy also reduces the end-to-end latency of a training round, from 90.5 ms to 75.2 ms in the clinical deployment, a reduction of 16.9 %. The benefit is consistent rather than incidental: it is positive in every simulated round, with a minimum of 8.2 % and a maximum of 25.8 %. It is, however, considerably smaller than the reduction in traffic, and the reason is visible in the decomposition: the communication term falls from 23.6 ms to 16.5 ms, while the compute terms — sensing, preprocessing, inference and aggregation — are common to every topology and dominate the total. A hierarchy relieves the link, not the computation, and a latency claim that did not separate the two would attribute to the topology an improvement it cannot produce.

The wearable deployment reduces round latency further, from 104.5 ms to 81.8 ms, or 21.7 %, with the communication term falling from 37.6 ms to 23.2 ms. That the larger topology benefits more is consistent with the mechanism rather than incidental to it: twenty-two stations contend for the same wireless medium where nine do not, so the latency attributable to the uplink is a larger share of the total and relieving it recovers more. The effect is in the direction the design predicts, and it is the reason the benefit should not be quoted as a single number divorced from the topology that produced it.

2) Cost of a valid guarantee: The results above were obtained under the oracle encoder configuration, in which

the frozen representation was pretrained on the pooled client training data. That configuration does not carry a valid end-to-end guarantee, and the utility it reports is consequently an upper bound. Re-running the proposed system under the disjoint configuration, in which the encoders see only a population reserved before federated training and disjoint from every client, costs 0.0322 in area under the ROC curve ($0.8429 \rightarrow 0.8107$), 0.0310 at the case level and 0.0763 in average precision.

That figure is the price of the guarantee rather than a deficiency of the method, and reporting it is the point: a system that pretrains on its own clients' data and reports a budget for the head alone has not made an end-to-end statement, and the difference quantifies exactly how much of the apparent performance depended on that shortcut. The comparisons between arms are unaffected, since every arm shares the same encoders; what changes is the level at which all of them sit.

3) Privacy and utility: The composed budget of 0.970 satisfies the $\varepsilon \leq 1$ requirement. Against a non-private reference trained under the identical recipe, the change in balanced accuracy under that budget is -0.92 pp with a paired $95\,\%$ confidence interval of $[-2.82, +0.99]$ over five seeds. The interval contains zero, so the correct reading is that no accuracy cost is detectable at this sample size, not that privacy improved the model; the point estimate is negative but the experiment cannot resolve an effect of that magnitude. The requirement of at most 2 pp is met on either reading.

On the wearable deployment the same comparison yields 1.967 pp with an interval of $[1.760, 2.173]$ over fifteen seeds. The interval spans the 2 pp threshold, so this objective is *inconclusive*: the sample size cannot determine on which side of the requirement the true value lies. The seed count was fixed in advance, and it is left at fifteen deliberately — extending it until the interval falls on the favourable side of a threshold would convert a pre-registered test into a search, and the resulting verdict would carry no evidential weight. The point estimate lies below the threshold and the upper bound above it; both are reported.

4) Quality of service: The interesting question at inference time is not whether the model is fast enough. It is whether the machinery that carries the guarantees is affordable, because the standing objection to privacy-preserving learning in real-time settings is that its apparatus costs more than the response budget allows. Table IV answers that directly.

Constructing the conformal prediction set — the component that delivers the coverage guarantee — costs 0.030 ms of a 4.401 ms mean inference path, or $0.68\,\%$ of it, and 0.037 ms at the 95th percentile. The differentially private mechanism contributes nothing to this path at all, and it is worth being exact about why: its noise is applied to gradients during training, so it alters the parameters that are served but adds no operation to serving them. The guarantee is therefore not made cheap by engineering; it is free by construction, and the only guarantee-bearing component that must be paid for at inference is the conformal set, at under one percent of the budget. That is the property which makes the framework deployable where response time binds, and it holds independently of how fast the underlying network happens to be.

The absolute figures are 4.58 ms at the 95th percentile in the clinical deployment and 1.22 ms in the wearable one, against a 100 ms budget, both measured on the integer-quantised graph executed on a CPU provider. The twentyfold margin should be attributed where it belongs: to a compact fusion head executed in integer arithmetic, not to the hierarchy or to any privacy mechanism. The difference between the two deployments follows input dimensionality rather than anything in the framework. What the margin establishes is not speed but headroom — that a deployment may spend its latency budget on a larger backbone without the guarantees becoming the constraint.

Update delivery succeeds on $100.00\,\%$ of transmissions in the clinical deployment and $99.94\,\%$ in the wearable one, against a $99\,\%$ requirement. Both figures are measured on every round rather than extrapolated; under the earlier configuration the clinical quantity read $99.27\,\%$, of which nine rounds in ten were analytically composed. The wearable deployment does not reach perfect delivery — its per-round minimum is $99.15\,\%$ — which is expected where twenty-two stations contend for one medium, and is the more representative of the two figures.

5) Energy: Per-round energy falls from 1266.2 J to 350.8 J in the clinical deployment, a saving of $72.3\,\%$, and from 3086.2 J to 714.8 J in the wearable one, a saving of $76.8\,\%$, against a required $20\,\%$. Energy is attributed by tier rather than reported in aggregate, because a hierarchy relocates computation rather than eliminating it, and an aggregate figure would conceal a saving at one tier purchased by an increase at another. Fig. 7 shows why this matters. Introducing the edge tier *alone* does not reduce what the constrained device tier spends; it leaves that unchanged and adds edge energy on top, so the total rises from 1266 J to 1344 J and the hierarchy is, by itself, a net cost. The saving arises only in the configuration that also shortens what each device computes and transmits, which reduces the device term fivefold. A total reported without the decomposition would have credited the topology with a benefit the topology does not produce.

B. The Privacy Unit and the Budget

Two comparisons quantify what the design decisions of Section III-D cost. Both are paired by seed against an arm that differs only in the factor under test.

1) Cost of moving the unit from the record to the individual: Enforcing the guarantee over individuals rather than records reduces the area under the ROC curve by 0.0188, with a paired $95\,\%$ interval of $[-0.0255, -0.0120]$ over five seeds; the case-level figure is -0.0193 $[-0.0266, -0.0120]$ and average precision falls by 0.0783 $[-0.1046, -0.0520]$. Unlike the accuracy comparison against a non-private reference, these intervals exclude zero: the cost is small but it is real, and it is larger in average precision than in the ranking measure, which is what one expects when the minority class is the harder one to fit under a coarser unit of sampling.

The mechanism of that cost is visible in the construction rather than mysterious. Sampling patients rather than records

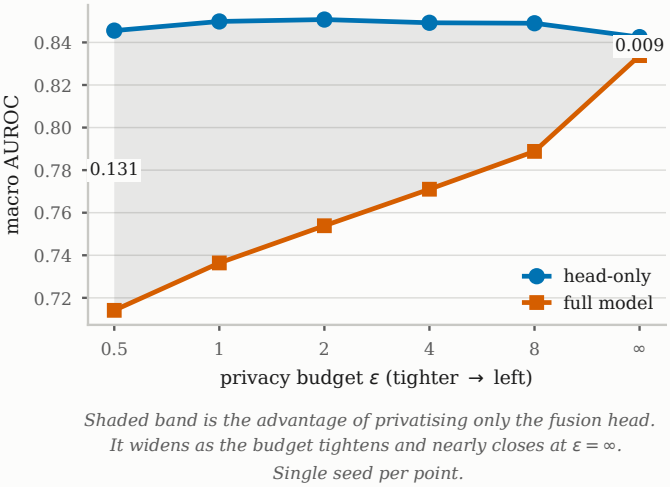


Fig. 8. Head-only against full-model private training across the budget. The advantage of privatising only the fusion head is negligible when no noise is added and grows monotonically as the budget tightens, which identifies it as a consequence of the privatised dimensionality rather than of a better starting point.

reduces the number of effective units per step from 34,044 to 4,372 on this partition, and bounding a patient's summed gradient constrains an individual contributing many records more tightly than one contributing few. Whether that second effect dominates — whether the cost grows with the number of records an individual contributes — is separated from the first in Section V-B3, where the number of units per step is held equal across the two arms so that only the clipping geometry differs.

2) Why the privatised subset is restricted to the head: Fig. 8 reports both recipes across the budget. Privatising only the fusion head is not merely a better initialisation: if it were, its advantage would be roughly constant in ε. It is not. At $\varepsilon = \infty$, where no noise is added, the two recipes are nearly indistinguishable, separated by 0.009; the gap grows monotonically as the budget tightens, reaching 0.060 at $\varepsilon = 8$ and 0.131 at $\varepsilon = 0.5$. The head-only curve is essentially flat across the whole range, moving by under 0.008 from $\varepsilon = 0.5$ to $\varepsilon = \infty$, while the full-model curve loses 0.119 over the same interval.

That pattern is the signature of a dimensionality effect rather than an optimisation one. The perturbation a parameter vector must absorb scales with its size, so restricting the privatised subset to $30\,\%$ of the network buys nothing when there is no perturbation and buys most when the perturbation is largest. It also means the recipe should not be recommended on the basis of non-private performance, where it is worth almost nothing.

These points are single-seed, and the monotonicity rather than any individual value is what the claim rests on; the head-only and full-model arms at $\varepsilon = 1$ are separately confirmed over five seeds in the baseline comparison.

3) Records per individual: The comparison above prices the unit at one operating point. Because a patient-level guarantee bounds the *summed* contribution of an individual, that price should not be a constant: it should grow with the number of records each individual contributes, and it should vanish when that number is one, since a patient who contributes a single record is a record. Table V reports both arms as k, the number of records retained per patient in training, is varied over $\{1, 2, 4, 8\}$ at three seeds. The number of units drawn per step is held equal in the two arms, so the sampling-rate effect identified above is removed by construction and only the clipping geometry differs. Validation and test partitions are untouched.

TABLE V
AVERAGE PRECISION BY RECORDS PER INDIVIDUAL k. THE GAP IS PAIRED BY SEED AND REPORTED AS MEAN $\pm$ THE HALF-WIDTH OF A STUDENT-t INTERVAL OVER THREE SEEDS; AN INTERVAL EXCLUDING ZERO IS ONE WHOSE HALF-WIDTH IS SMALLER THAN ITS MEAN.

k	Record	Individual	Gap	p
1	0.5015	0.5067	-0.0052 ± 0.0613	0.7514
2	0.5590	0.5084	0.0507 ± 0.0090	0.0017
4	0.5924	0.5092	0.0832 ± 0.0138	0.0015
8	0.5955	0.5089	0.0866 ± 0.0263	0.0049

At $k = 1$ the two arms are indistinguishable: the gap is -0.0052 with an interval covering zero. This is the prediction the construction makes, and it is the reason the remaining rows can be read as an effect of the unit rather than of the harness — an apparatus that separated the arms at $k = 1$ would be separating them for some reason other than the guarantee. From $k = 2$ the gap is resolved and grows monotonically, reaching 0.0866 at $k = 8$ with the increment between $k = 4$ and $k = 8$ an order of magnitude smaller than the increment before it.

The direction of the divergence is worth stating precisely, because the aggregate gap conceals it. The individual-level arm is flat in k, moving within $[0.4967, 0.5176]$ across every configuration, while the record-level arm rises from 0.5015 to 0.5955. The widening gap is therefore not degradation under the stricter unit but the absence of improvement: additional records per patient are additional signal that the record-level arm converts into accuracy and the individual-level arm cannot, because a bound on the summed contribution tightens on exactly the individuals who supply the extra records. A practitioner reading this should not expect an individual-level guarantee to become cheaper by collecting more data per patient; on this corpus it becomes more expensive, and it does so by forgoing what the extra data would otherwise buy.

The ranking measure moves in the same direction — 0.0006, 0.0094, 0.0133 and 0.0132 over the same k — but every interval covers zero at three seeds, so the effect is resolved in average precision only and the AUROC trend is reported as consistent rather than established. Accuracy is not reported for these arms: it is pinned near the majority rate throughout by an uncalibrated decision threshold on an imbalanced corpus, and an area under a curve of approximately 0.835 confirms the models rank well despite it.

C. Does Fusion Earn Its Place?

The framework assumes two modalities are worth combining. That assumption is tested directly, and separately from the federated system, because the question concerns the data

TABLE VI
COMPONENT ABLATION ON THE CLINICAL DEPLOYMENT. ΔAUROC IS PAIRED AGAINST THE FULL SYSTEM BY SEED OVER THREE SEEDS, REPORTED AS MEAN ± THE HALF-WIDTH OF A STUDENT-t INTERVAL; AN INTERVAL EXCLUDES ZERO WHEN THE HALF-WIDTH IS SMALLER THAN THE MEAN, AND THE ONE THAT DOES IS SET IN BOLD. MB IS THE TOTAL COMPRESSED PAYLOAD UPLOADED ACROSS TRAINING. C6 IS AN *inclusion* ARM: FOR IT A RISE, NOT A FALL, WOULD INDICATE A CONTRIBUTION.

Removed	AUROC	ΔAUROC	MB
— (full)	0.8541	—	252
C1 hierarchy	0.8484	-0.0058 ± 0.0088	50
C2 privacy	0.8307	$\mathbf{-0.0234 \pm 0.0205}$	252
C3 sparsification	0.8553	$+0.0012 \pm 0.0083$	1260
C4 quantisation	0.8543	$+0.0001 \pm 0.0006$	403
+C6 ASWA	0.8542	$+0.0001 \pm 0.0009$	252

rather than the protocol: federation and privacy noise would add variance and confounds to a comparison about whether two streams carry complementary information. Each arm is retrained from scratch on its own inputs rather than evaluated with a modality zeroed, since feeding zeros to a jointly trained model measures what happens when a stream fails, not what it was worth.

Over five seeds the fused model attains 0.8653 ± 0.0072 in area under the ROC curve against 0.8283 ± 0.0087 for the better single modality and 0.7680 ± 0.0045 for the weaker, each interval a Student-t half-width as elsewhere in this section, a gain of 0.0370 and 5.9 times the seed standard deviation; in average precision the gain is 0.0671. The modalities are associated by the recording apparatus — the same patient during the same operation — so the gain is not an artefact of pairing rule.

The seed count matters here and is reported rather than buried. At two seeds the same experiment returned 0.0411 and 0.1062, at 14.1 times the seed standard deviation; three further seeds reduced the average-precision figure by $37\,\%$ and the effect size by more than half. The conclusion is unchanged and the five-seed figures are the ones quoted, but the shrinkage is a reminder that an effect measured on two seeds is a hypothesis rather than a result.

The wearable deployment cannot corroborate this and the reason is a property of that task. Its better single modality already attains 0.9976, leaving a fusion gain of 0.0004 and no headroom in which a second stream could demonstrate anything. Reporting this matters more than omitting it would: multimodal federated learning is not universally motivated, and on a task one sensor already solves the second adds cost without information. The case for fusion is made per task, and the presence of two sensors is not itself the case.

D. Necessity of the Components

Each component is removed in turn from the full system, every other setting held fixed, and the arms are paired by seed over three seeds. Table VI reports the result. Read on accuracy alone the table is largely empty, and that is the finding rather than a shortfall: of the four removals only the privacy mechanism produces an interval excluding zero.

1) Compression is justified by traffic, not by accuracy: Removing top-κ sparsification changes accuracy by $+0.0012$ with an interval spanning zero, and multiplies uplink by 5.0. Removing quantisation changes accuracy by $+0.0001$ — an interval of width 0.0011, the tightest in the study — and multiplies uplink by 1.60. Neither component earns its place by what it does to the model; both earn it by what they do to the link, and together they account for an eightfold reduction in payload at no measurable cost in discrimination. An ablation reported only in the accuracy column would have concluded that both were redundant.

2) The hierarchy is not an accuracy mechanism: Collapsing the edge tier moves AUROC by -0.0058 with an interval covering zero. This is the expected result and not a negative one: the hierarchy is introduced to relieve the uplink and the energy budget, quantities reported in Section V-A, and it would be surprising if a change of aggregation topology altered what the model can discriminate. The uplink figure for this arm is not comparable with the others and is reported only for completeness, because collapsing the tier also reduces the number of edge rounds per global round from five to one; the traffic difference is an artefact of that schedule rather than of the topology.

3) Removing the privacy mechanism does not recover utility: The one interval that excludes zero belongs to the component expected to cost the most. Training the same system without differential privacy *reduces* area under the ROC curve by 0.0234 $[-0.0439, -0.0029]$, and inflates the seed-to-seed spread in average precision from ± 0.014 to ± 0.070. The mechanism is visible in the checkpoint selection: two of the three non-private seeds attain their best validation score at the first round of ten and degrade thereafter, where every private seed peaks between rounds six and nine. Clipping each contribution and adding calibrated noise evidently regularises a compact head trained over frozen features, and removing that regularisation lets the head overfit almost immediately.

This should be read narrowly. The comparison prices the privacy mechanism against a baseline that carries no substitute regularisation beyond checkpoint selection, not against a non-private system tuned for generalisation; a practitioner who removed the mechanism would presumably reintroduce weight decay or early stopping in its place. The defensible claim is therefore that removing the guarantee did not recover utility on this task, not that differential privacy improves accuracy in general. It does establish that the accuracy cost of the guarantee, which the objective of Table III bounds at two points, is not being paid at this operating point.

4) An inclusion arm that earned its removal: The aggregation rule inherited from the earlier design of this framework was never measured. Added to the full system it moves AUROC by $+0.0001$, with an interval of width 0.0018 that covers zero, and leaves uplink unchanged. Since this is an inclusion rather than an ablation, the sign convention inverts and a rise would have indicated a contribution; none is observed. The component is therefore reported as measured and not adopted, which is a stronger statement than omitting it would have been.

TABLE VII
PATIENT-LEVEL MEMBERSHIP INFERENCE. MEAN ATTACK AUC OVER SEEDS WITH 95 % INTERVALS; 0.5 IS CHANCE. THE CONTROL IS NON-PRIVATE TRAINING OF THE SAME HEAD OVER THE SAME FROZEN ENCODERS, AND IS A PRECONDITION OF THE AUDIT RATHER THAN A SUPPLEMENTARY EXPERIMENT.

Configuration	Attack	AUC	95 % CI
Control (non-private)	lik. ratio	0.6230	[0.598, 0.648]
Record-level DP	lik. ratio	0.5477	[0.488, 0.608]
Patient-level DP	threshold	0.5224	[0.518, 0.527]

E. Membership-Inference Audit

The guarantee reported above is an upper bound on the influence of an individual. What an adversary attains in practice is a separate question, and Table VII reports it. Three configurations were attacked under both attacks of Section III-J, each with sixteen shadow models trained on independently drawn halves of the patient population, and each scored on the same 600 member and 600 non-member patients.

The control establishes that the audit is capable of detecting leakage. On non-private training of the same head over the same frozen encoders the likelihood-ratio attack reaches 0.6230, exceeding the 0.5711 obtained there by loss thresholding: per-patient calibration recovers 0.0519 of signal that a global threshold leaves buried in per-patient difficulty, which is precisely the confound illustrated in Fig. 5. Every negative result below is conditioned on this — the same procedure, applied to a configuration known to memorise, finds it.

The control is insensitive to how long it trains, which is worth recording because it bounds what the configuration can establish. Repeating it at ten rounds rather than one moves the likelihood-ratio attack by -0.0047 and no attack of the four by more than 0.006. A head of 560k parameters over frozen encoders evidently reaches its memorisation capacity almost immediately, so the control's strength is a property of the architecture rather than of the training schedule. This matters for interpretation in one direction only: it means the control cannot be strengthened by training it longer, and a leakage larger than the one reported here would have to be sought with a different configuration rather than a longer run.

Applied to the private configurations the same procedure finds very little. Record-level differential privacy yields 0.5477 with an interval that includes chance, and patient-level differential privacy 0.5224. The comparison the experiment was designed to make is the paired difference between them, computed over five seeds on identical shadow splits, and it is not significant under any of the four attacks: $+0.0138$ $[-0.0034, +0.0310]$ for the loss threshold, $+0.0325$ $[-0.0280, +0.0929]$ for the offline likelihood ratio, $+0.0034$ for the online variant and -0.0211 for its pooled-variance form. Every interval spans zero.

Two readings must be separated. The *predicted* effect is refuted: the hypothesis under test was that a record-level guarantee would be exploitable at the level of the individual, with an attack approaching 0.7 against roughly 0.5 for a patient-level guarantee, and an effect of that size lies far outside every interval obtained. A *small* effect is not excluded: at five seeds the half-width on the strongest attack is ± 0.060, three of the four attacks point in the predicted direction, and the loss threshold reaches $p = 0.090$. The defensible statement is that the large effect predicted does not exist, and that an effect below roughly six points of AUC cannot be resolved at this sample size. It would not be defensible to state that the two guarantees are equivalent.

One diagnostic supports the reading that the defence is working rather than that the attack is failing. On the patient-level arm the likelihood ratio is *worse* than the plain loss threshold, by 0.0071. Per-patient calibration helps only where there is difficulty-confounded signal to recover; that it helps on the control and not here is what a defence leaving no such signal looks like. True-positive rates at 1 % false-positive rate are correspondingly low throughout, at 0.029 on the control and 0.010 on the patient-level arm. At 0.1 % the rate is undefined: with 600 non-members the receiver operating characteristic cannot resolve that false-positive rate, and interpolating a value there would report an artefact of the curve rather than a measurement.

The consequence for the framework is narrower than the hypothesis intended, but it is not nothing. The analytic argument of Section III-D stands — a record-level budget bounds the wrong quantity, and the conversion to an individual-level one is prohibitive — and the utility cost of enforcing the individual-level guarantee directly is small. What the audit does not supply is a demonstration that the weaker guarantee is exploitable in practice, on this data, at this budget. Reporting that plainly, alongside a control showing the attack works where leakage exists, is what makes the result interpretable rather than merely absent.

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